

Deep Learning Model Performance Comparison  
Neural Network Architecture Benchmark

| Model Name              | Parameters (M) | Training Time (h) | Inference Time (ms) | ImageNet Accuracy (%) | ImageNet mAP (%) | COCO mAP (%) | Memory (GB) |
|-------------------------|----------------|-------------------|---------------------|-----------------------|------------------|--------------|-------------|
| ResNet-50               | 25.5           | 48.2              | 1250                | 76.1                  | 92.9             | 37.2         | 2.1         |
| ResNet-101              | 44.5           | 72.5              | 890                 | 77.8                  | 93.7             | 39.1         | 3.2         |
| EfficientNet-B0         | 5.3            | 36.1              | 2150                | 77.1                  | 93.5             | 38.4         | 0.9         |
| EfficientNet-B3         | 19.3           | 98.5              | 1050                | 82.9                  | 96.3             | 42.8         | 2.8         |
| Vision Transformer-B/16 | 86.6           | 156               | 850                 | 81.1                  | 95.3             | 40.5         | 4.2         |
| YOLO-v8n                | 3.2            | 24                | 3500                | 72.5                  | 90.1             | 35.6         | 0.6         |
| YOLO-v8s                | 11.2           | 42                | 2800                | 75.2                  | 91.8             | 37.9         | 1.2         |
| YOLO-v8m                | 25.9           | 68                | 1950                | 79.3                  | 94.1             | 40.2         | 2.3         |
| Swin-Transformer-TL     | 28.3           | 120               | 720                 | 81.5                  | 95.5             | 41.1         | 3.5         |
| ConvNeXt-L              | 28.6           | 110               | 950                 | 82.1                  | 95.9             | 41.8         | 3.3         |

## Model Performance Analysis

| Model       | Params (M) | Accuracy (%) | Efficiency (FLOPs) | Params-per-Speed Ratio | Accuracy (Memory per Accuracy) |
|-------------|------------|--------------|--------------------|------------------------|--------------------------------|
| ResNet-50   | 25.5       | 76.1         | 2.984314           | 0.977011               | 0 1.465517                     |
| ResNet-10   | 44.5       | 77.8         | 1.748315           | 1.600719               | 1.7 2.401079                   |
| EfficientNe | 5.3        | 77.1         | 14.54717           | 0.195572               | 1 0.293358                     |
| EfficientNe | 19.3       | 82.9         | 4.295337           | 0.586626               | 6.8 0.879939                   |
| Vision Tra  | 86.6       | 81.1         | 0.93649            | 2.784566               | 5 4.176849                     |
| YOLO-v8n    | 3.2        | 72.5         | 22.65625           | 0.142222               | -3.6 0.213333                  |
| YOLO-v8s    | 11.2       | 75.2         | 6.714286           | 0.444444               | -0.9 0.666667                  |
| YOLO-v8m    | 25.9       | 79.3         | 3.061776           | 0.883959               | 3.2 1.325939                   |
| Swin-Tran:  | 28.3       | 81.5         | 2.879859           | 0.898413               | 5.4 1.347619                   |
| ConvNeXt    | 28.6       | 82.1         | 2.870629           | 0.890966               | 6 1.336449                     |

## Training Progress - Loss and Accuracy

| Epoch | Training L | Validation | Training A | Validation | Learning R | Overfitting Gap (Val-Train) |
|-------|------------|------------|------------|------------|------------|-----------------------------|
| 1     | 2.302      | 2.298      | 12.5       | 15.2       | 0.001      | -0.004                      |
| 2     | 1.956      | 1.845      | 28.3       | 35.1       | 0.001      | -0.111                      |
| 3     | 1.542      | 1.423      | 42.7       | 48.9       | 0.001      | -0.119                      |
| 5     | 0.892      | 0.934      | 65.3       | 62.1       | 0.001      | 0.042                       |
| 10    | 0.345      | 0.412      | 87.2       | 84.5       | 0.0005     | 0.067                       |
| 20    | 0.156      | 0.198      | 94.2       | 91.3       | 0.0001     | 0.042                       |
| 30    | 0.089      | 0.127      | 96.5       | 93.8       | 0.00005    | 0.038                       |
| 40    | 0.052      | 0.098      | 97.8       | 94.9       | 0.00001    | 0.046                       |
| 50    | 0.031      | 0.084      | 98.6       | 95.5       | 0.000001   | 0.053                       |

## Hyperparameter Tuning Results

| Experiment | Learning Rate | Batch Size | Weight Decay | Warmup Steps | Final Accuracy | Converger | Accuracy per Hour |
|------------|---------------|------------|--------------|--------------|----------------|-----------|-------------------|
| 1          | 0.001         | 32         | 0.0001       | 500          | 91.2           | 48        | 1.9               |
| 2          | 0.001         | 64         | 0.0001       | 500          | 92.1           | 45        | 2.046667          |
| 3          | 0.001         | 128        | 0.0001       | 500          | 91.8           | 42        | 2.185714          |
| 4          | 0.002         | 64         | 0.0001       | 1000         | 93.5           | 46        | 2.032609          |
| 5          | 0.002         | 64         | 0.00005      | 1000         | 93.2           | 47        | 1.982979          |
| 6          | 0.0005        | 64         | 0.0001       | 500          | 92.8           | 52        | 1.784615          |
| 7          | 0.002         | 128        | 0.0001       | 1000         | 94.1           | 44        | 2.138636          |
| 8          | 0.002         | 128        | 0.00005      | 1500         | 94.8           | 48.5      | 1.954639          |